Certificate in Healthcare Infection Prevention & Control (CHIP) Internship

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Organization and Institutional Linkages

My internship placement will be at the Center for Research and Innovations in Special Populations (CRIISP) in the Division of Geriatric and Palliative Medicine at University of Michigan, under the supervision of Dr. Lona Mody as a research assistant. The CRIISP applies various translational epidemiologic research methods from observational and molecular epidemiology, clinical trials to enhance disease outcomes in vulnerable populations. As a part of the Healthcare Infection Prevention and Control (CHIP) program at the UM School of Public Health, this internship provides the knowledge and skills needed to work in this field including surveillance, clinical microbiology, and infectious disease epidemiology.

Internship Objectives and Responsibilities

Internship Objectives:

1. Obtain knowledge about surveillance for antibiotic resistant organisms, infection prevention & control guidelines and research protocols
2. Experience "real world" application of infection prevention & control policy/practice in post-acute care settings including the development of collaborative, multidisciplinary relationships
3. Develop individual research project based on current studies including: hypothesis, data entry and analysis, laboratory analysis, and write up
4. Write an abstract on individual research projects and submit to at least one national meeting for the opportunity to present. Take a lead in developing their presentation with support and mentorship from CRIISP staff.

I assisted in multiple areas of the CRIISP projects including:

1. Assist in laboratory with microbiology testing and cultures;
2. Data entry and analysis of MDRO colonization, clinical data, antibiotic usage, and/or other Infection Control topics
3. Analyze MDRO colonization and contamination rates, assessing for trends and improvements, and providing monthly feedback reports to facilities
4. Assist in visits to post-acute care facilities with the field staff, including patient recruitment, specimen collection, medical record review and educational activities
5. Independent project development, including literature reviews, generating and testing hypotheses, creating tables and figures, and abstract and paper writing.

Projects Description

Background

The persistence of multidrug-resistant organisms (MDROs) in long-term care facilities (LTCFs) presents a significant challenge to infection control efforts. Among the most concerning MDROs are multidrug-resistant gram-negative bacteria (RGNB), methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant *enterococci* (VRE).

Among residents of the long-term care facilities, the average duration of MDRO colonization was approximately 3 months[1]. Previous studies have reported that the median duration of RGNB colonization in LTCFs is approximately 144 days, with a range of 41 to 349 days [2]. Notably, clearance of RGNB colonization was observed in only 39% of colonization episodes, with a clearance rate of 2.6 episodes per 1000 days, indicating that colonization often persists for extended periods in this population[2]. Similarly, MRSA colonization poses significant challenges due to its ability to persist in patients over time. In the case of VRE, the duration of colonization is also prolonged[3].

Various risk factors have been associated with the prolonged persistence of MDROs in LTCFs. For example, patients with liver disease, presence of comorbidities, a history of infection within the previous year, or those utilizing permanent urinary catheters, nasogastric tubes, open wounds, or gastrostomy before hospitalization have been shown to have significantly increased odds of prolonged MDRO colonization[4–7]. Mattner et al. (2010) identified key factors influencing the persistence of MRSA, including age, gender, decolonization therapy, and the extent of anatomical site colonization[8]. Patients with colonization at multiple sites exhibited a significantly higher risk of persistent colonization, with a hazard ratio of 2.18 (95% CI: 1.52–3.15, $p < 0.01$), highlighting the importance of the infection's anatomical spread in determining persistence[8]. The prolonged VRE duration is also related to antibiotic use at admission and the presence of comorbid conditions such as pressure ulcers[3]. These findings underscore the multifaceted nature of MDRO persistence, where both patient-related and device-related factors play critical roles.

Finally, the distinction between persistent and intermittent carriage of MDROs is vital in understanding the dynamics of colonization. This differentiation is crucial for epidemiological studies and for tailoring infection control measures in LTCFs, where MDRO colonization can serve as a reservoir for transmission to other vulnerable patients.

Understanding these factors and the duration of colonization is essential for informing targeted interventions to reduce the burden of MDROs in LTCFs and improve patient outcomes.

Objectives

1. Describe persistence of pre-existing colonization and new MDRO (RGNB, MRSA, VRE) acquisitions. Identify risk factors associated with persistence.
2. Describe MDRO clearance. Clearance and time to clearance will be assessed for the subset of patients with MDRO persistence.

Methods

Study Population and Design

We conducted a prospective cohort study of recently admitted NF patients between November 2013 and May 2016. Patients from 6 NFs in southeast Michigan were enrolled within 14 days of NF admission and followed for up to 6 months. We evaluated MDRO colonization prevalence upon enrollment, any new acquisitions and spontaneous losses of MDROs during their stay, and MDRO prevalence on discharge. We confirmed dates and discharge destination in the minimum data set [18]. Patients receiving end-of-life care were excluded. A written informed consent by the patient or his/her proxy was required. This research was approved by the University of Michigan Institutional Review Board.

Clinical Data Collection

Demographic data (including age, sex, race) and risk factors for MDRO colonization (including functional disability, use of indwelling devices, presence of comorbidities, prior antibiotic use, and length of hospitalization) were collected by trained research personnel for each patient. Functional disability was measured using the Physical Self-Maintenance Scale (PSMS), which ranges from 6 (full independence) to 30 (full dependence) in 6 categories of self-maintenance (bathing, dressing, feeding, ambulation, grooming, and toileting [19, 20]). We also tested specific disabilities in dressing (at least moderate assistance), bathing (unable to independently bathe more than hands and face), toileting (soiling or wetting more than once a week), or grooming (regularly needs at least moderate assistance or supervision). Device use was defined as having a feeding tube or indwelling urinary catheter in place at the time of enrollment. Prolonged hospital stay prior to the post–acute care stay was ≥14 days. The Charlson comorbidity score [21] was calculated, and high comorbidity was defined as a score of ≥3. Prior documented antibiotic use was defined as any antibiotic use within the past 30 days, including prior to NF admission.

Microbiologic Methods

Microbiological samples were collected from patients' hands, nares, oropharynx, enteral feeding tube insertion site, suprapubic catheter site, groin, perianal area, and wounds to assess MDRO colonization on the day of enrollment, day 14, day 30, and then monthly for up to 6 months. Samples were collected using sterile swabs in transport media (BactiSwab, Remel, Lenexa, Kansas) and cultured on mannitol salt agar, MacConkey agar, and bile-esculin plates with 6 µg/mL vancomycin. Hand swabs were enriched overnight in brain heart infusion broth at 36°C before culturing. Staphylococcus aureus, vancomycin-resistant enterococci (VRE), and gram-negative bacilli (GNB) were identified using standard microbiologic testing described previously [12]. Antimicrobial susceptibility testing for S. aureus and GNB isolates was performed by disk diffusion per Clinical and Laboratory Standards Institute guidelines, including screening for extended-spectrum β-lactamases (ESBLs) [22]. For S. aureus, methicillin resistance was determined by disk diffusion method using 30-µg cefoxitin disks. For GNB, the following antibiotics were tested: ceftazidime (30 µg), ceftazidime/clavulanic acid (30/10 µg), ciprofloxacin (5 µg), and imipenem (10 µg); resistance to 1 or more these was considered resistant GNB (R-GNB). We identified the species of VRE (ie, Enterococcus faecalis, Enterococcus faecium, other enterococci) using the multiplex polymerase chain reaction assay [23]. Results of microbiological testing were not disclosed to NF staff or administrators during the study period.

Statistical Analysis

Persistent carriage is characterized by two or more positive cultures separated by fewer than two negative cultures, while intermittent carriage involves two or more negative cultures following a single positive culture[9].

To evaluate factors potentially associated with persistent MRSA, VRE, RGNB carriage, for univariate analysis, continuous variables were compared using the nonparametric Wilcoxon rank-sum test and t-test, and categorical variables were compared using Pearson's chi-square test or Fisher's exact test, as appropriate. ORs and 95% CIs were calculated. Next, Multivariate analysis using conditional logistics regression was performed, in which variables were statistically significant (P=.1) in univariate models were entered into the multivariable model. All tests were 2-sided, and P values <.05 were considered significant. Statistical analysis was performed with use of R version 4.4.0 (RStudio, PBC, Boston, MA, USA)[1].

Duration of persistence was analyzed using similar methods, with tests selected based on the assumptions of normality and equal variances, including Welch's t-test, Mann-Whitney U test, One-Way ANOVA, Welch's ANOVA, or Kruskal-Wallis test for multiple-level categorical predictors.The duration of clearance of MRSA, VRE, RGNB, and any MDRO among NH residents were described using Kaplan-Meier estimates and actuarial life-table methods[2].

Variables (Potential Risk Factors)

Variable	Variable Name (field name)	Description
Age	age_bl	Resident age, in years
Sex	sex_bl	Resident sex, 1=male 0=female
Race	race_bl	resident race, 1=white 2=black or african american 3=asian 4=american indian or alaska native 5=native hawaiian or other pacific islander 99=not reported or unknown
Ethnicity	ethnic_bl	resident ethnicity, 1=hispanic or latino 2=not hispanic or latino 99=not reported or unknown
BMI	BMI	BMI score
Intervention	intervention	1=Intervention facility; 0=Control facility
Antibiotic Use	antibiotic_bl	Patient receiving antibiotics at baseline, 1=Yes; 0=No
Feeding Tube Use	feedtube	Feeding Tube Use: 1=yes, 0=no, 99=unknown
Urinary Catheter Use	urincath	Urinary Catheter Use: 1=yes, 0=no, 99=unknown
Device use	deviceuse	Device Use: 1=yes, 0=no, 99=unknown
Wound	openwound_final	Wounds Present: 1=yes, 0=no, 99=unknown
Cognitive Sum Score	cogsum_bl	Cognitive status, sum score
History of MRSA	homrsa_bl	Resident history of MRSA colonization/infection, 1=yes 0=no 99=unknown
History of VRE	hovre_bl	Resident history of VRE colonization/infection, 1=yes 0=no 99=unknown

History of R-GNB	horgnb_bl	Resident history of Resistant gram-negative bacilli colonization/infection, 1=yes 0=no 99=unknown
Functional disability (PSMS Total Score)	psmscore	PSMS Total Score
Length of Hospitalization	hospstay	Patient's length of hospital stay prior to NH admission
Charlson Comorbidity Score (Number)	charlson	Patient's Charlson Comorbidity Score
Charlson Comorbidity Score (Category)	charlson_cat	Charlson Comorbidity Score - 1, 2, 3, 4, >=5
Myocardial Infarction	ccsmi_bl	Charlson's comorbidity score - myocardial infarction 1=yes 0=no 99=unknown
Congestive Heart Failure	ccschf_bl	Charlson's comorbidity score - congestive heart failure 1=yes 0=no 99=unknown
Peripheral Vascular Dx	ccspvod_bl	Charlson's comorbidity score - peripheral vascular occlusive disease 1=yes 0=no 99=unknown
Cerebral Vascular Accident	ccscva_bl	Charlson's comorbidity score - cerebral vascular accident 1=yes 0=no 99=unknown
Dementia	ccsdemen_bl	Charlson's comorbidity score - dementia 1=yes 0=no 99=unknown
COPD	ccscopd_bl	Charlson's comorbidity score - Chronic obstructive pulmonary disease 1=yes 0=no 99=unknown
Connective Tissue Dx	ccscontis_bl	Charlson's comorbidity score - connective tissue disease 1=yes 0=no 99=unknown
Peptic Ulcer Dx	ccspud_bl	Charlson's comorbidity score - peptic ulcer disease 1=yes 0=no 99=unknown
Mild Liver Dx	ccsmld_bl	Charlson's comorbidity score - mild liver disease 1=yes 0=no 99=unknown
Diabetes	ccsdm_bl	Charlson's comorbidity score - diabetes mellitus 1=yes 0=no 99=unknown
Hemiplegia	ccshemi_bl	Charlson's comorbidity score - hemiplegia 1=yes 0=no 99=unknown
Mod-Severe Renal Dx	ccsmsrd_bl	Charlson's comorbidity score - Moderate/severe renal disease 1=yes 0=no 99=unknown
Diabetes End Organ Dx	ccsdmeod_bl	Charlson's comorbidity score - diabetes mellitus with end organ disease 1=yes 0=no 99=unknown
Any Tumor	ccstumor_bl	Charlson's comorbidity score - any tumor 1=yes 0=no 99=unknown
Leukemia	ccsleuk_bl	Charlson's comorbidity score - leukemia 1=yes 0=no 99=unknown
Lymphoma	ccslymph_bl	Charlson's comorbidity score - lymphoma 1=yes 0=no 99=unknown
Mod-Severe Liver Dx	ccsmsld_bl	Charlson's comorbidity score - Moderate/severe liver disease

		1=yes 0=no 99=unknown
Metastatic Solid Tumor	ccsmst_bl	Charlson's comorbidity score - metastatic solid tumor 1=yes 0=no 99=unknown
AIDS	ccsaids_bl	Charlson's comorbidity score - AIDS 1=yes 0=no 99=unknown
Site Colonized	RGNBsiteX	Site X positive for rgnb, 1=Yes; 0=No
Site Colonized	MRSAsiteX	Site X positive for mrsa, 1=Yes; 0=No
Site Colonized	VREsiteX	Site X positive for vre, 1=Yes; 0=No
Multiple Body Sites	gnbr_p	Number of patient sites positive for gnbr
Multiple Body Sites	mrsa_p	Number of patient sites positive for mrsa
Multiple Body Sites	vre_p	Number of patient sites positive for vre
Room Surfaces	gnbr_env	Number of environmental (E or Eq) sites positive for gnrb
Room Surfaces	mrsa_env	Number of environmental (E or Eq) sites positive for mrsa
Room Surfaces	vre_env	Number of environmental (E or Eq) sites positive for vre
Room Surfaces (Category)	X_env_cat	Number of environmental (E or Eq) sites positive for X - 0, 1, 2, 3, >=4
Multiple Body Sites (Category)	X_p_cat	Charlson Comorbidity Score - 0, 1, 2, >=3

Reference

1. Lin IW, Huang CY, Pan SC, Chen YC, Li CM. Duration of colonization with and risk factors for prolonged carriage of multidrug resistant organisms among residents in long-term care facilities. *Antimicrobial Resistance & Infection Control*. 2017;6(1):86. doi:10.1186/s13756-017-0246-4
2. O'Fallon E, Gautam S, D'Agata EMC. Colonization with multidrug-resistant gram-negative bacteria: prolonged duration and frequent cocolonization. *Clin Infect Dis*. 2009;48(10):1375-1381. doi:10.1086/598194
3. Sohn KM, Peck KR, Joo EJ, et al. Duration of colonization and risk factors for prolonged carriage of vancomycin-resistant enterococci after discharge from the hospital. *Int J Infect Dis*. 2013;17(4):e240-246. doi:10.1016/j.ijid.2012.09.019
4. Lucet JC, Koulenti D, Zahar JR. Persistence of colonisation with MDRO following discharge from the ICU. *Intensive Care Med*. 2014;40(4):603-605. doi:10.1007/s00134-014-3258-z
5. Robicsek A, Beaumont JL, Peterson LR. Duration of Colonization with Methicillin-Resistant Staphylococcus aureus. *Clinical Infectious Diseases*. 2009;48(7):910-913. doi:10.1086/597296
6. Scanvic A, Denic L, Gaillon S, Giry P, Andremont A, Lucet JC. Duration of Colonization by Methicillin-Resistant Staphylococcus aureus after Hospital Discharge and Risk Factors for Prolonged Carriage. *Clinical Infectious Diseases*. 2001;32(10):1393-1398. doi:10.1086/320151
7. Diego Velez J, Orrego M, Montes S, et al. 541. Factors Associated with the Persistence of Colonization by Multidrug-Resistant Organisms in Cali, Colombia. *Open Forum Infect Dis*. 2019;6(Suppl 2):S258. doi:10.1093/ofid/ofz360.610
8. Mattner F, Biertz F, Ziesing S, Gastmeier P, Chaberny IF. Long-term persistence of MRSA in re-admitted patients. *Infection*. 2010;38(5):363-371. doi:10.1007/s15010-010-0038-8

Persistence of Newly Acquired Organisms_Project Plan_Skylar Zeng

Projected Benefits for Academic Program and Future Career

Academic Program:

This internship at CRIISP provides an invaluable opportunity to apply the knowledge I gained from courses in my first year study in SPH including Infectious Diseases, Hospital Epidemiology, Investigate Public Health Issues, and Epidemiological Data Analysis to practical healthcare settings. By assisting in ongoing projects, I am expecting to translate my understanding of infectious disease dynamics into actionable strategies for infection prevention, integrating data analysis techniques to enhance surveillance and response, and applying evidence-based approaches to address real-world challenges in the field.

Moreover, this internship aligns with several MPH Foundational Competencies, including applying epidemiological methods to real-world health settings, selecting appropriate data collection methods, analyzing data using biostatistics and informatics tools, interpreting analytical results, and working effectively in interprofessional teams alongside patients, hospital staff, and researchers. The internship also directly maps to the Hospital and Molecular Epidemiology (HME) competencies, enabling me to demonstrate sterile techniques and lab safety measures, evaluate molecular tools for outbreak investigations, interpret molecular epidemiological study results, analyze the role of epidemiologists in healthcare settings, and apply principles of antibiotic stewardship.

This comprehensive experience will not only solidify my mastery of the core competencies outlined above but also help me identify any academic knowledge gaps in real-world research settings, thereby informing my course selection and study methods for the next year to enhance my preparedness further.

Future Career:

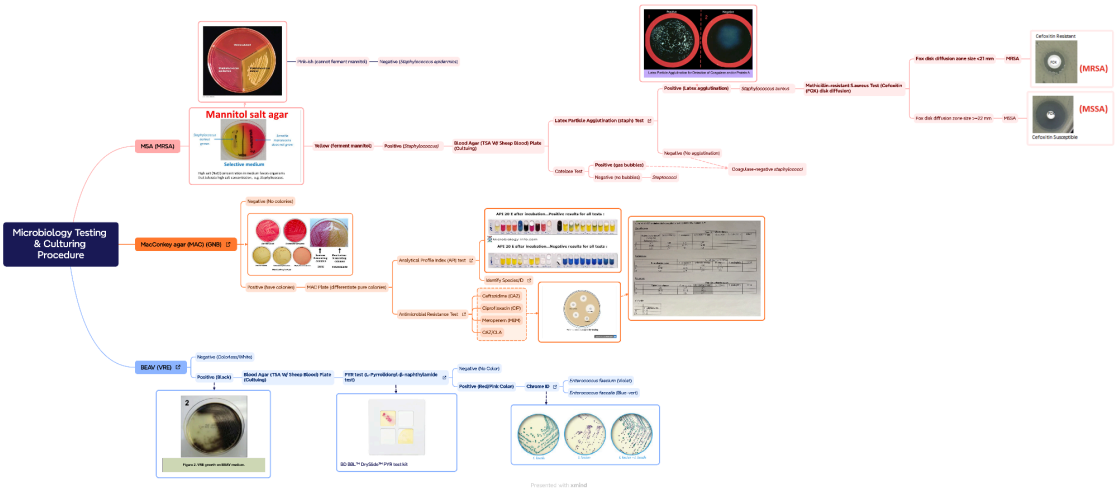
This internship provides a pivotal opportunity to propel my career aspirations of becoming an infection preventionist (IP). By immersing myself in the dynamic research environment at the CRIISP, I will gain invaluable exposure to the infection prevention and control research field and comprehensive understanding of the practical applications and real-world challenges faced in this domain, which is essential for completing the Associate - Infection Prevention and Control (a-IPC) and the CHIP certificate.

Collaborating with experienced professionals, actively participating in ongoing projects, and designing individual projects will allow me to explore infection prevention and control issues from a multifaceted perspective. I will have the chance to delve into various aspects, such as surveillance strategies, outbreak investigations, antimicrobial resistance mitigation, and the implementation of evidence-based interventions. Also, I seek to refine my skills in analyzing and interpreting data from laboratory and medical records and performance improvement, which are essential components of an IP’s role.

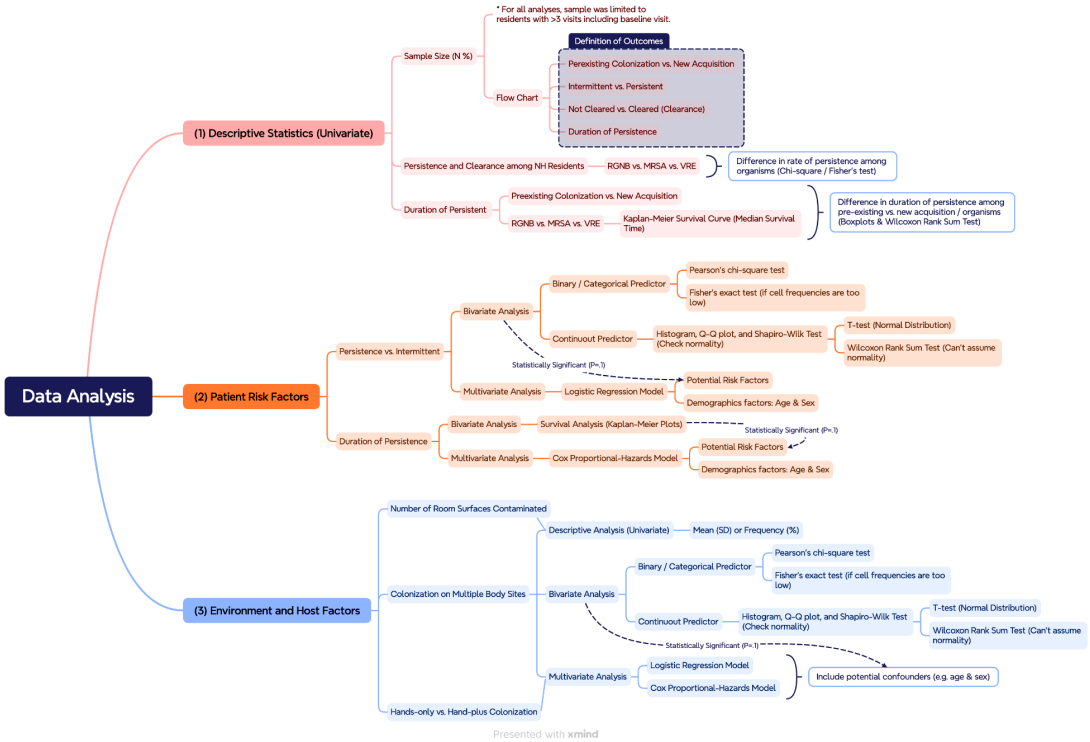
Moreover, this internship will expose me to the complicated dynamics of healthcare delivery systems (especially in the specific population), enabling me to comprehend the interaction between clinical and non-clinical services in ensuring patient safety and public health. Learning from diverse professionals will provide invaluable insights into the operational processes, decision-making frameworks, and the collaborative efforts required to effectively address infection prevention and control challenges within diverse healthcare settings.

Related Files:

- VA Merit Review Microbiology Testing Process:



- Data Analysis Process:



- Proposal and Responsibilities: CHIP Internship Proposal
- Project: Persistence of Newly Acquired Organisms_Project Plan_Skylar Zeng
- Website: https://criisp-mody.lab.medicine.umich.edu/home